

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/E_DEB_SAALYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3330, C2 = 6666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

#f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.19524100	290.00300	0.28459800	0.20273800	0.92453800	0.46387800

2.5%	0.14197100	270.05300	0.19406300	0.17056700	0.87551370	0.34177000
97.5%	0.44352770	317.54200	0.32704500	0.23715315	0.93186155	0.53973700
Min	0.12409100	90.58550	0.15393000	0.13630100	0.01107090	0.15520200
Max	0.73223300	328.49200	0.87744000	0.26890400	0.94240200	0.59267700
Result_mean	0.23649170	293.28982	0.26433027	0.20347863	0.91076787	0.44463164
Result_sd	0.08513385	12.68564	0.03446162	0.01707895	0.01517936	0.05066175
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	dv.1.		wE.1.
Result_mode	40.171500	12.299600	0.007321490	7.8371300	1.060880e-09	
2.5%	40.243200	10.799945	0.003391160	6.1314500	1.155000e-09	
97.5%	68.530990	17.061845	0.011879100	8.7111800	6.723712e-07	
Min	40.003000	9.258610	0.000125481	0.5220400	1.001200e-09	
Max	107.006000	32.418600	0.016585500	9.8683200	2.155700e-06	
Result_mean	48.243927	13.621597	0.007002104	7.4205124	9.125608e-08	
Result_sd	7.401431	1.622537	0.002334515	0.6595875	1.794059e-07	
	Sigma_W.1.					
Result_mode	0.17110700					
2.5%	0.15344105					
97.5%	0.19810700					
Min	0.13764700					
Max	5.67894000					
Result_mean	0.17470616					
Result_sd	0.04745068					

Number of observations, n = 277
 Number of parameters, k = 11
 Log-likelihood, LL = -78.31844
 BIC = 218.5011

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.09	1.28
E_coc.1.	1.16	1.46
Ehp.1.	1.07	1.19
Fm.1.	1.00	1.00
kap.1.	1.01	1.02
kapA.1.	1.14	1.42
pAm.1.	1.12	1.36
r_pAm_pM.1.	1.16	1.49
rOM_ClxHorse.1.	1.01	1.04
Sigma_W.1.	1.02	1.05

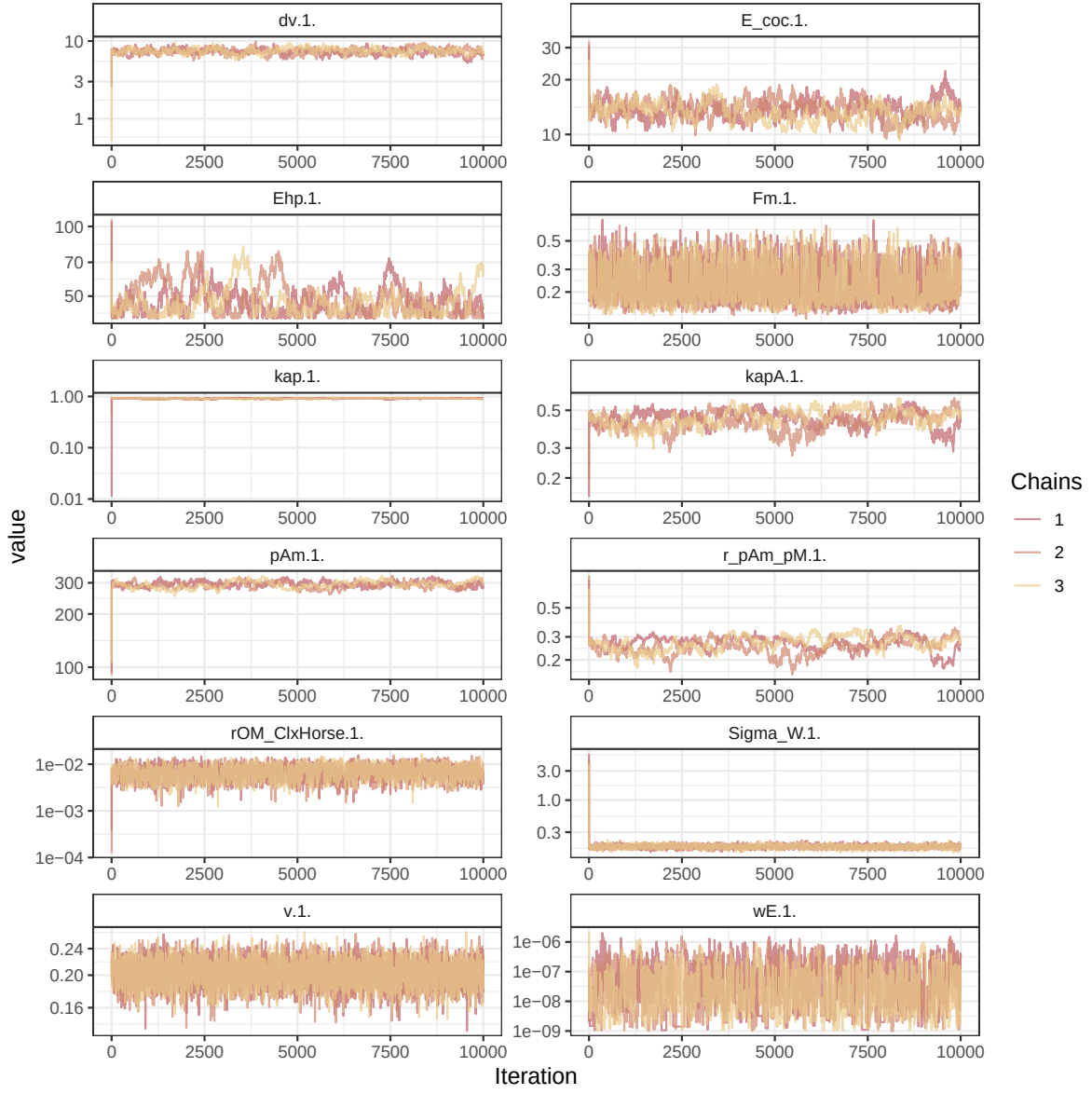


Figure 1: Parameters chains. Only the first iteration and the last r `Nb_iter_kept` are represented.

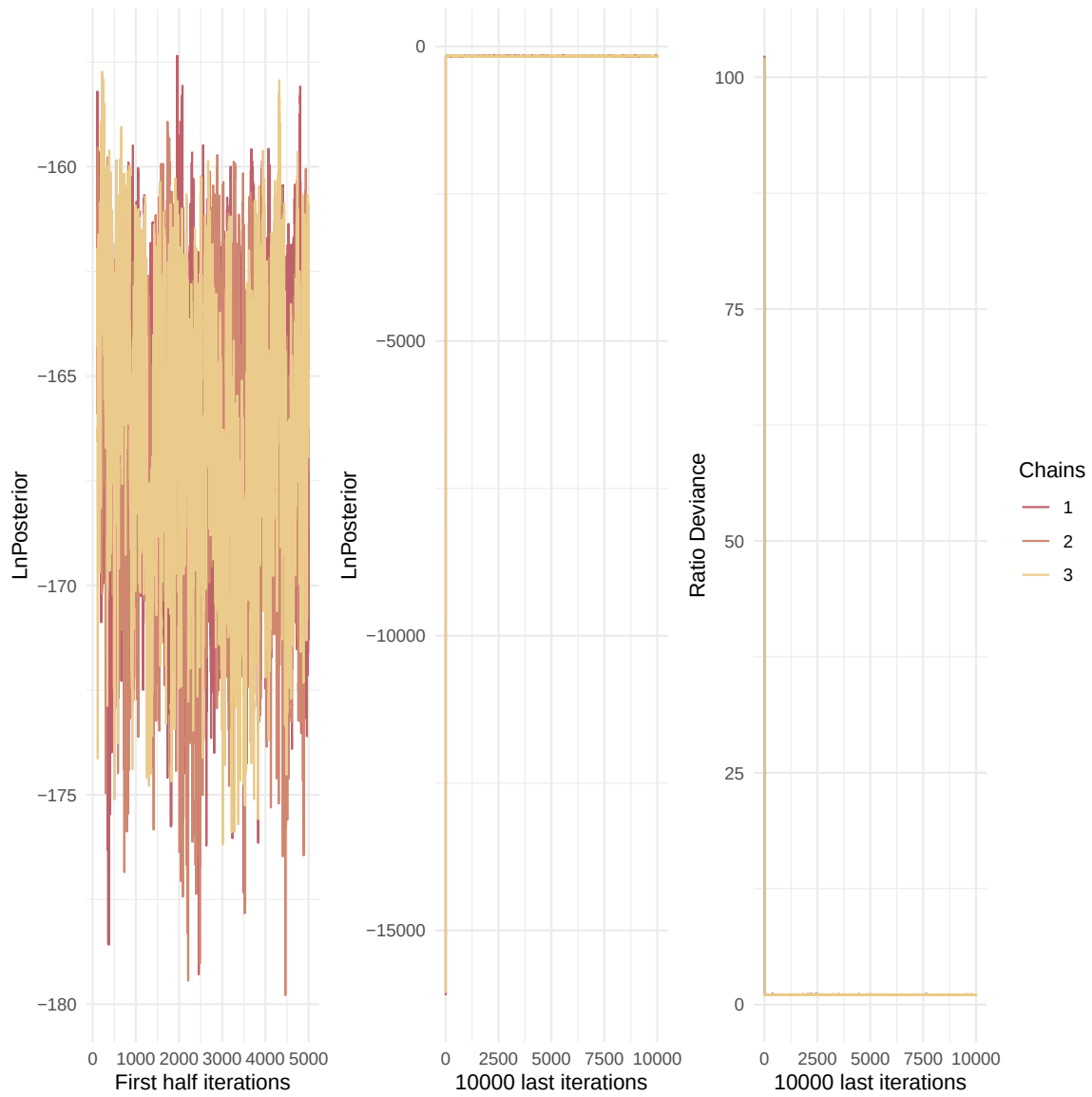


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

v.1.	1.00	1.00
wE.1.	1.04	1.09

Multivariate psrf

1.12

```
mcmc_list_corr <- mcmc.list(  
  lapply(mcmc_list, function(x) {  
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc  
  })  
)  
  
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))  
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

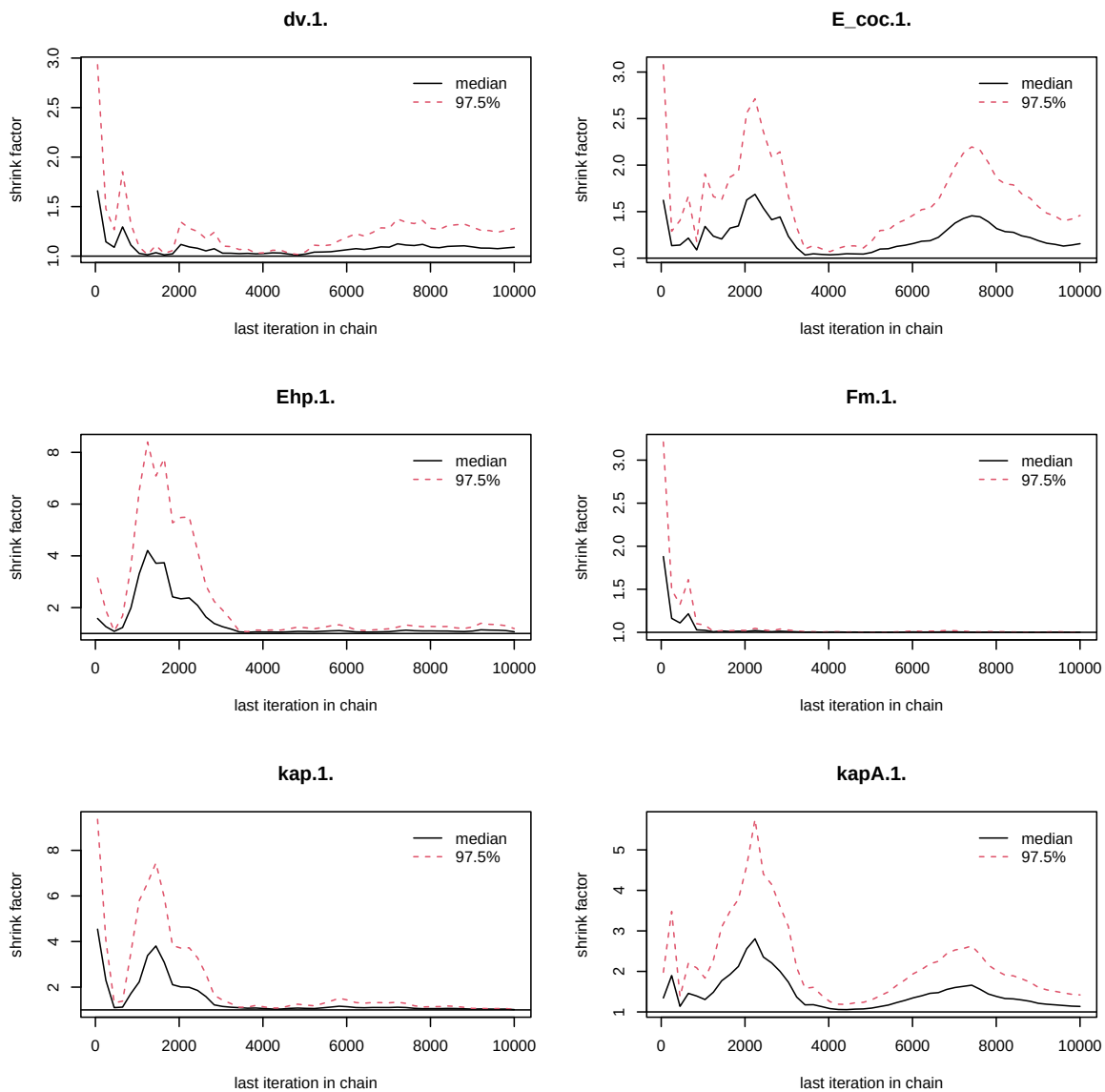


Figure 3: Chains convergence.

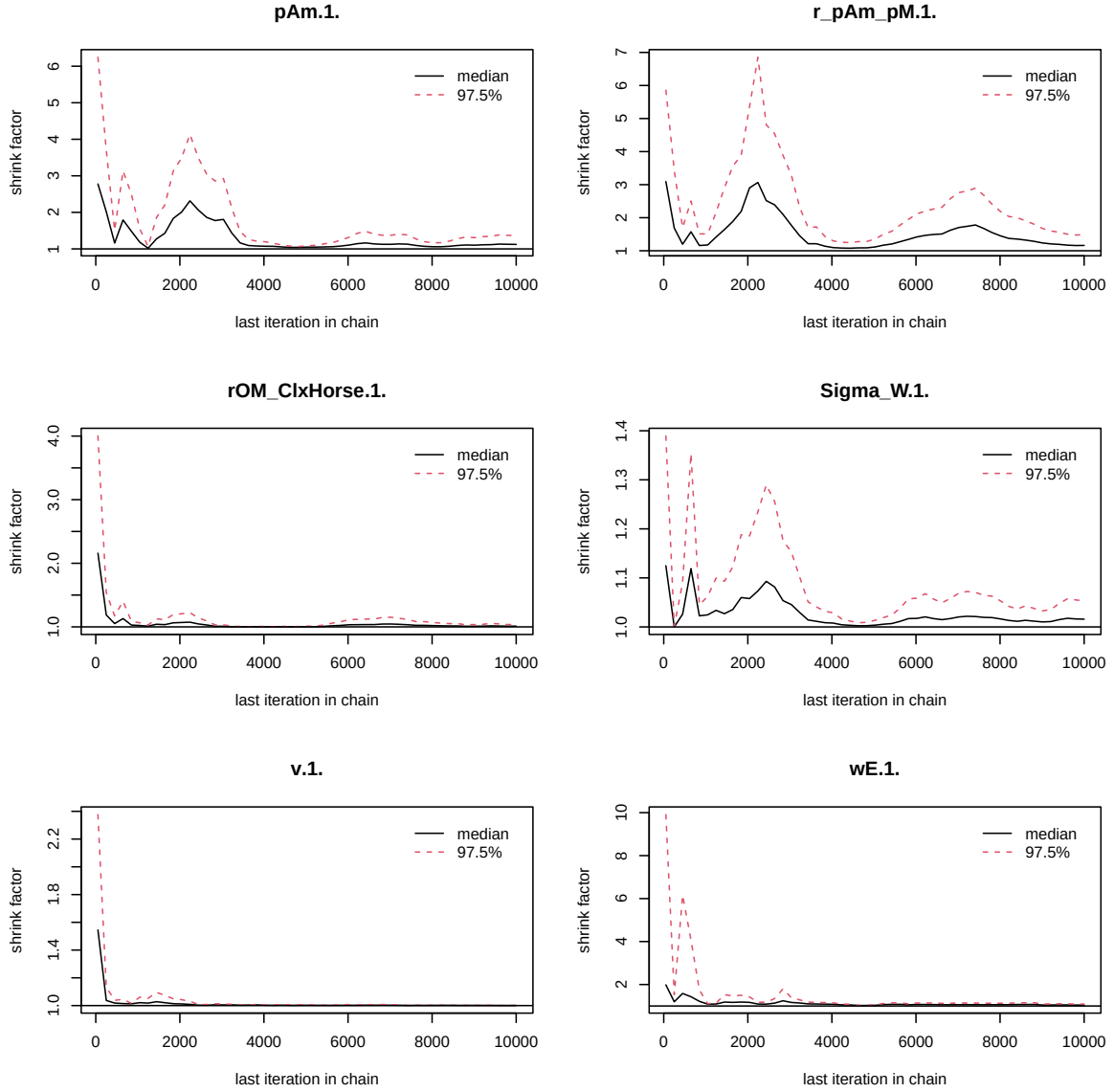


Figure 4: Chains convergence.

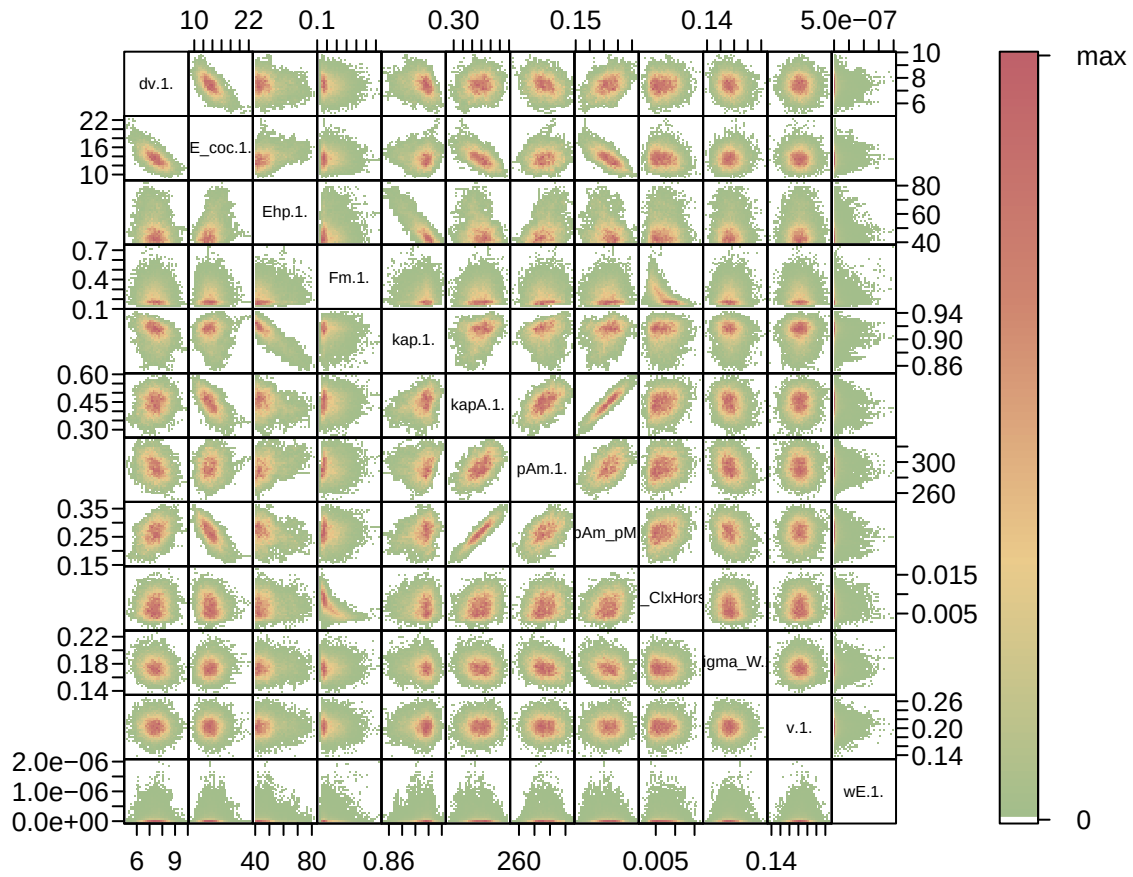


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

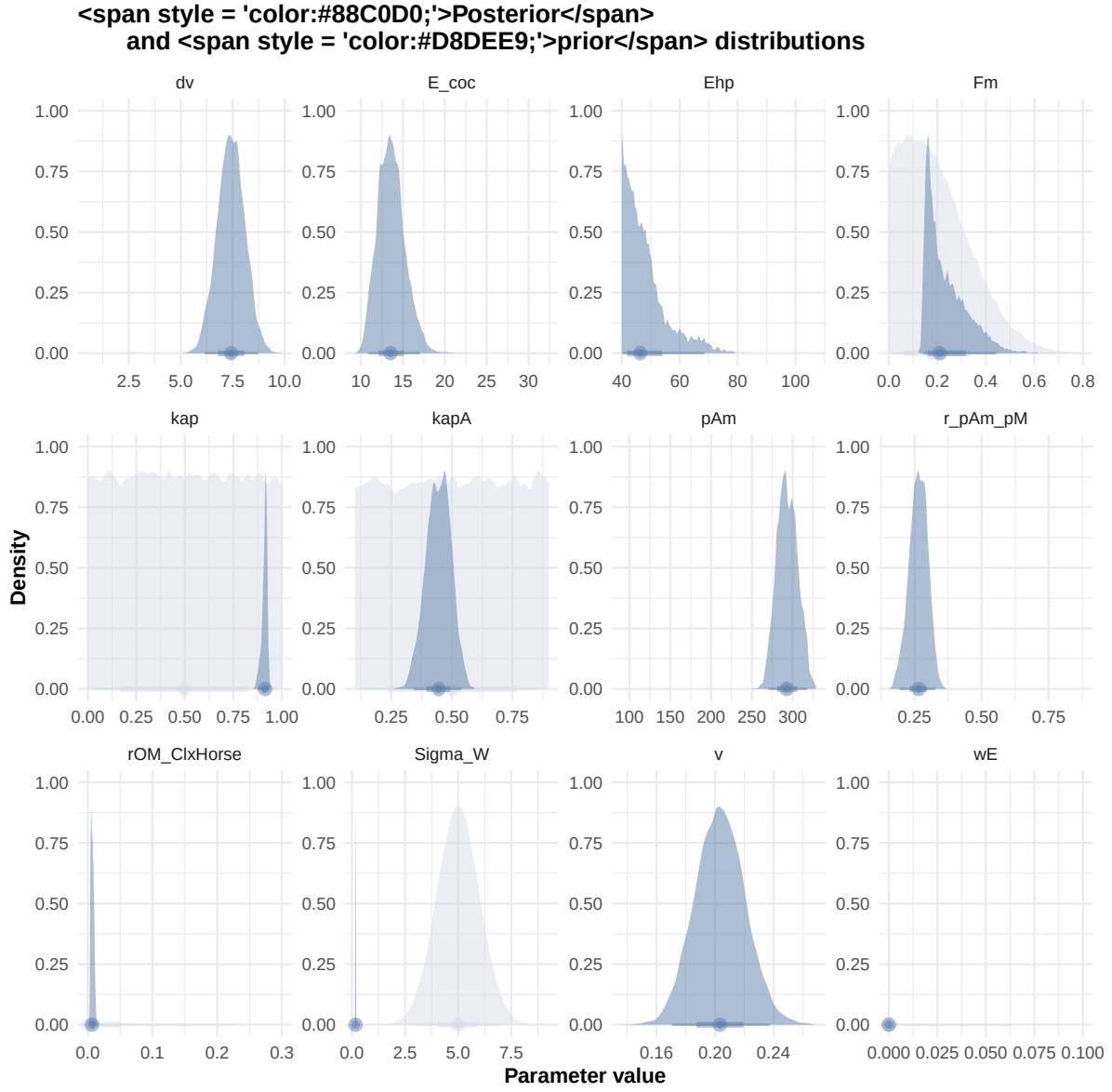


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

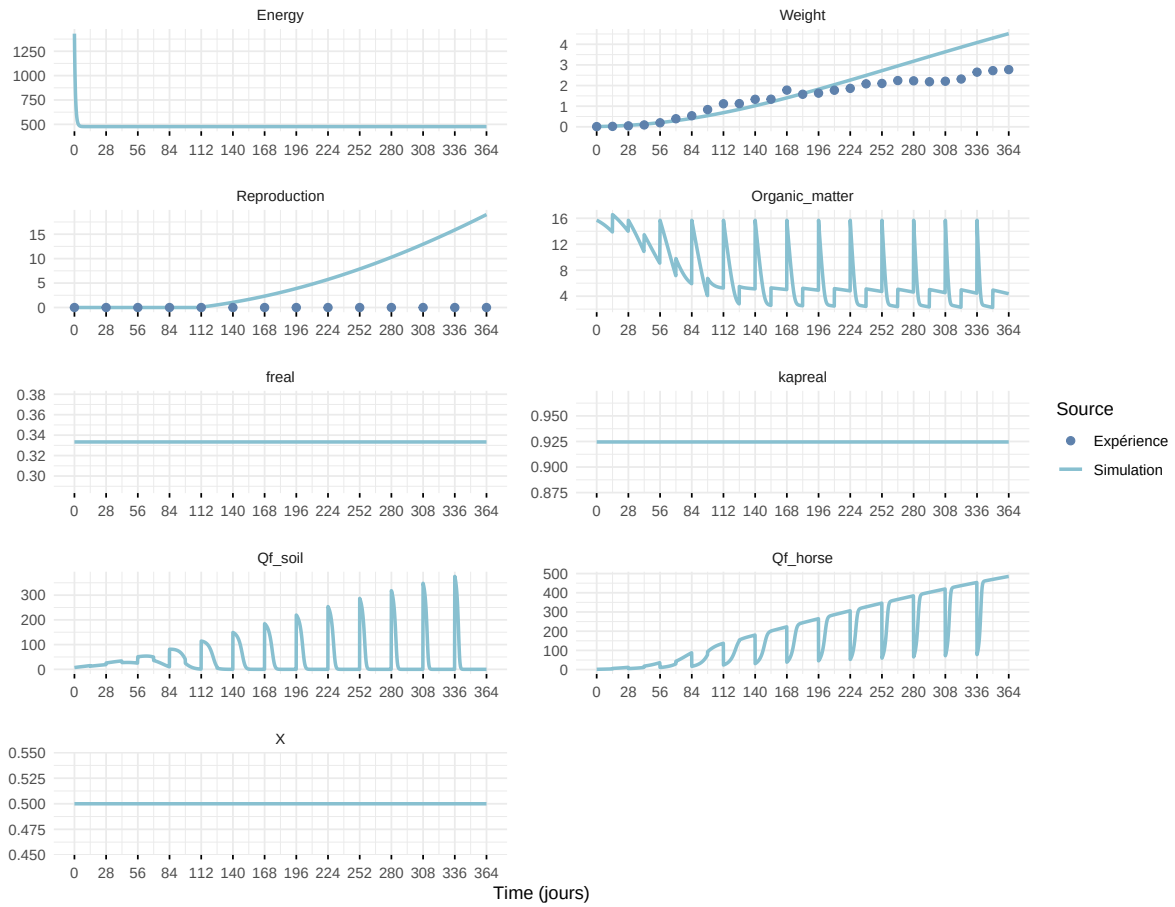
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



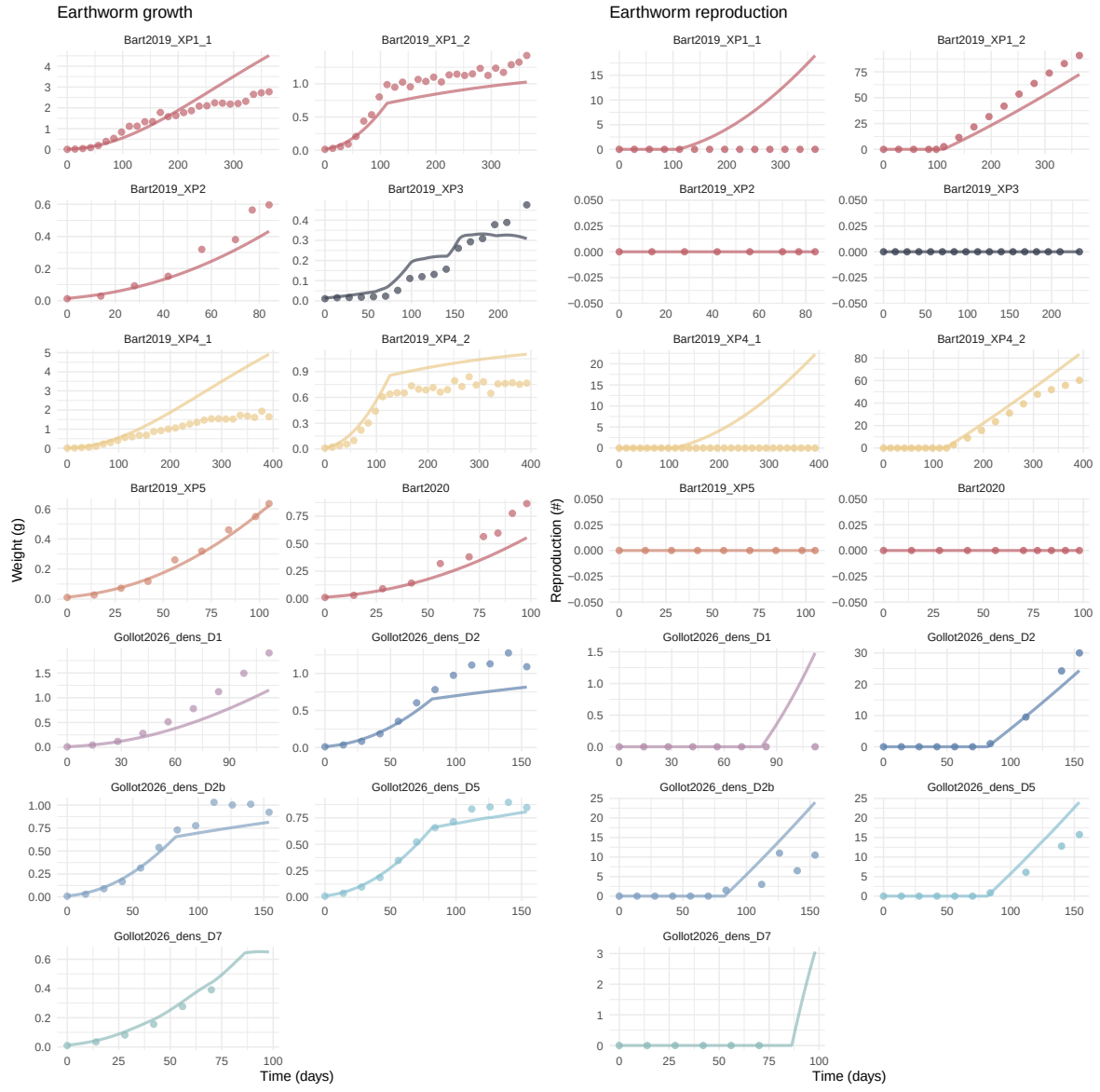
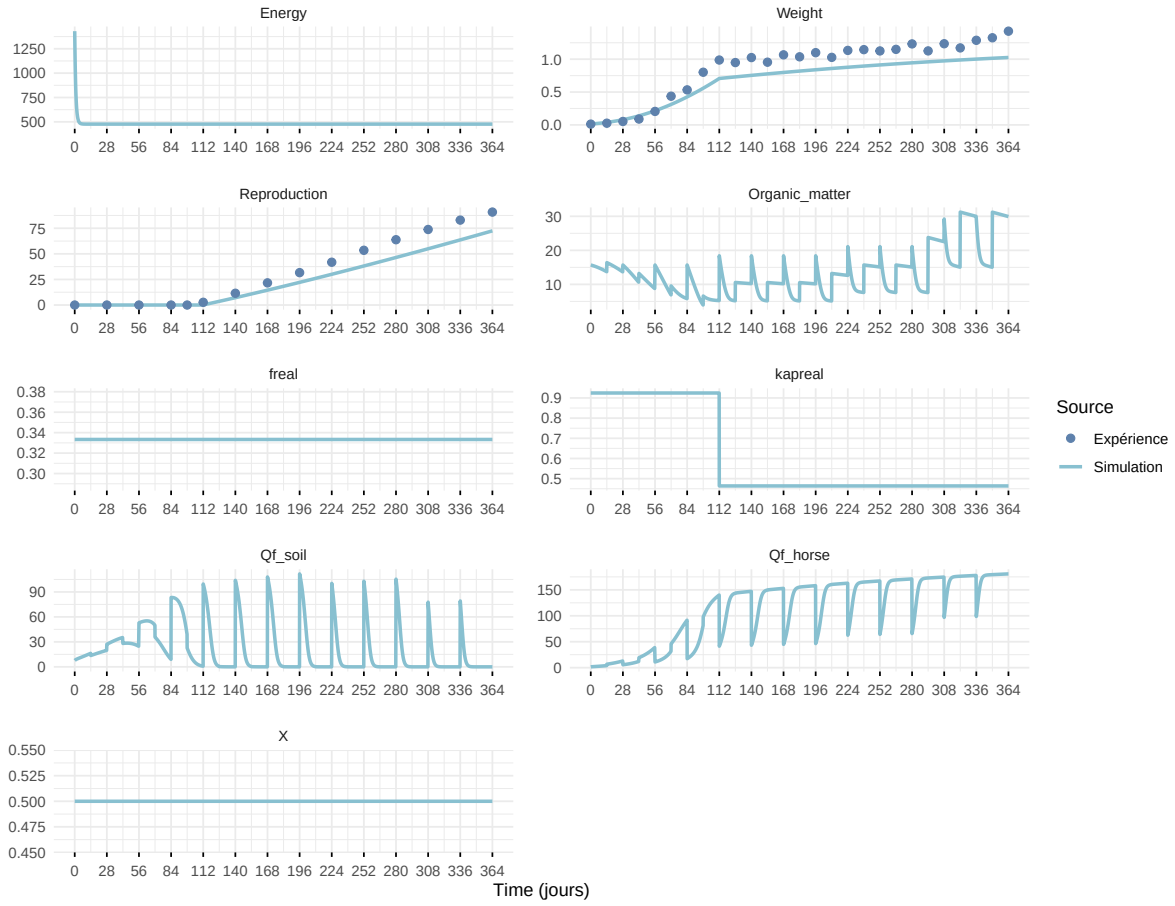
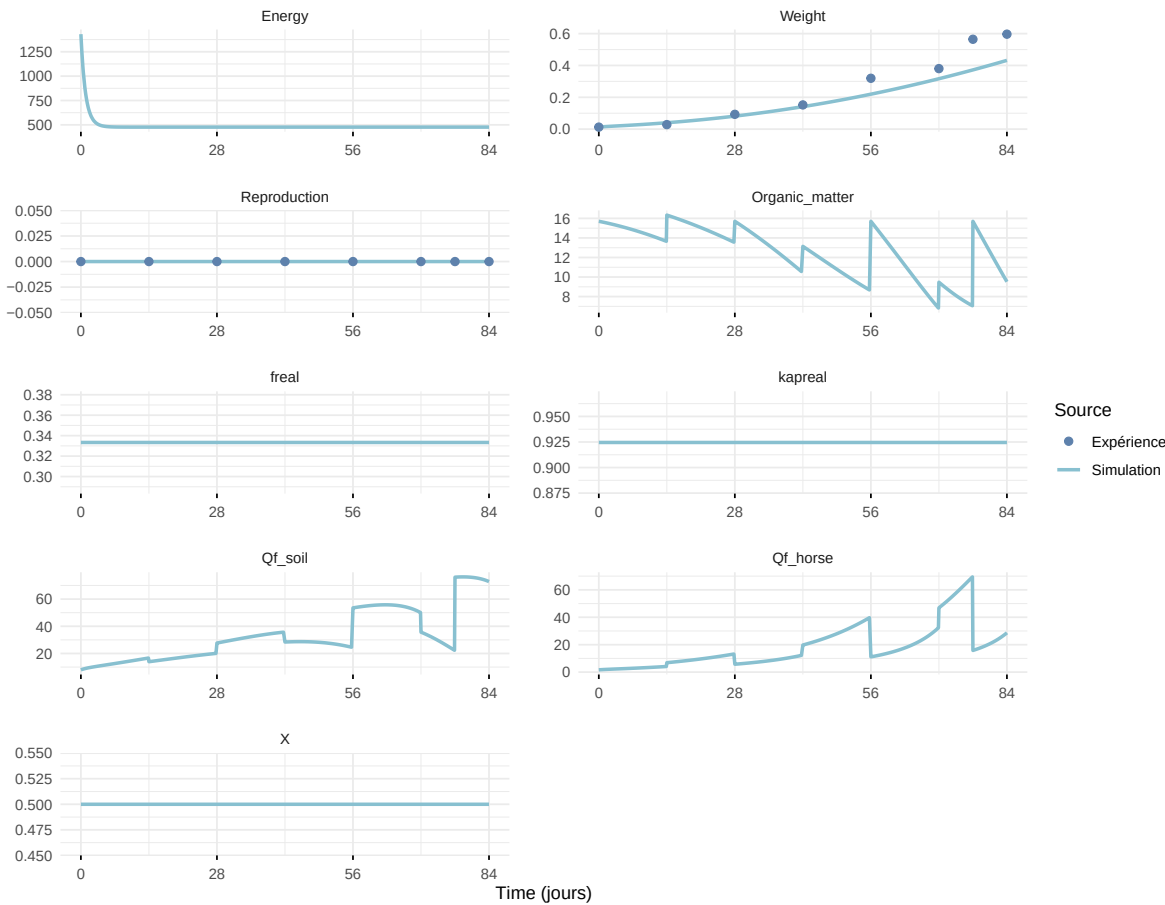


Figure 7: Simulations Weight and Reproduction.

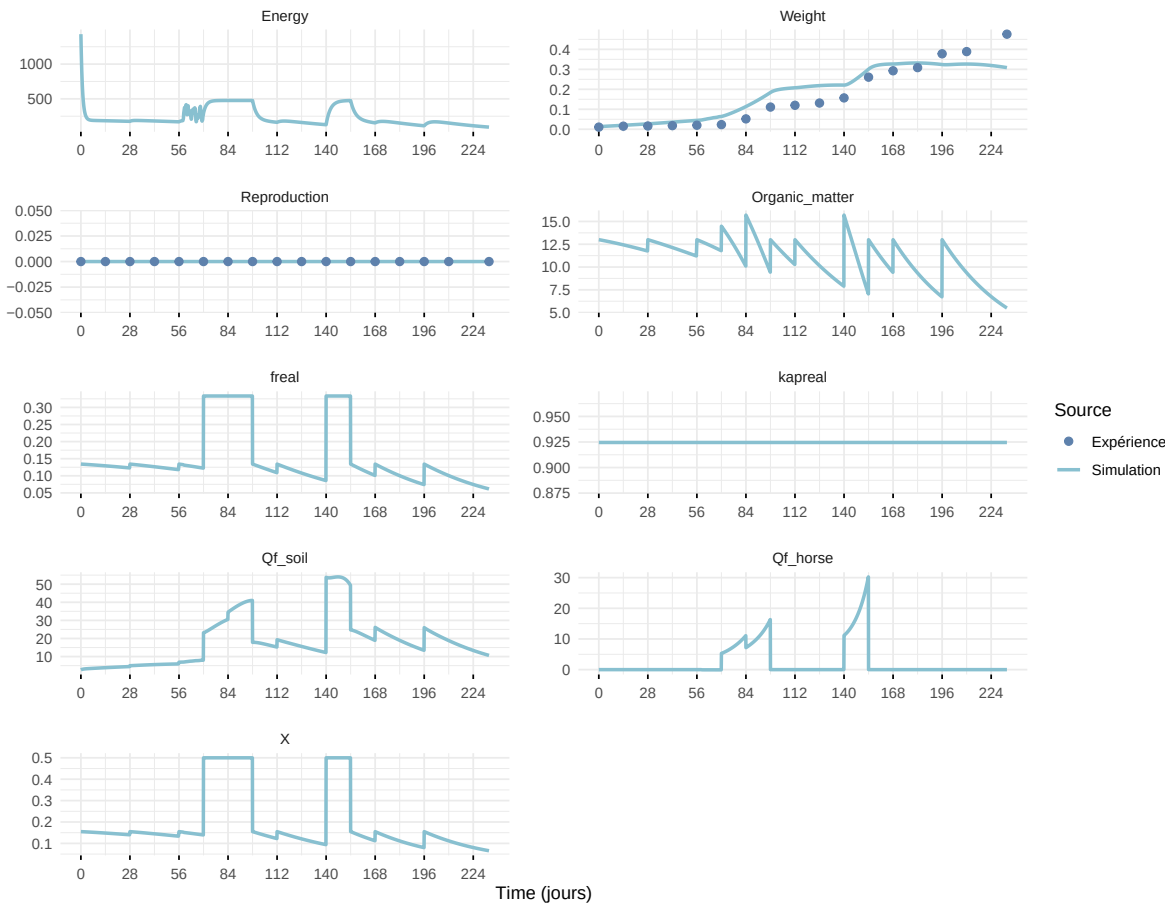
Expérience : Bart2019_XP1_2



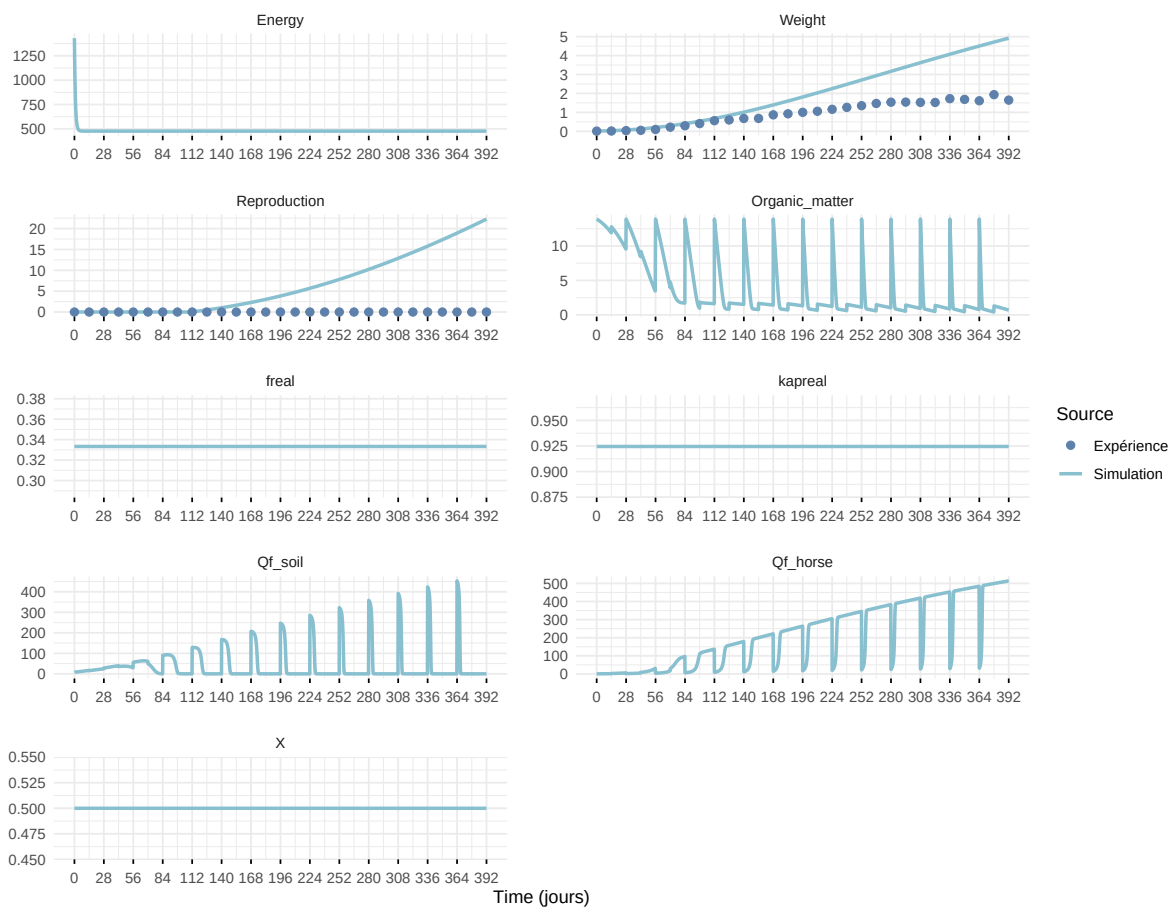
Expérience : Bart2019_XP2



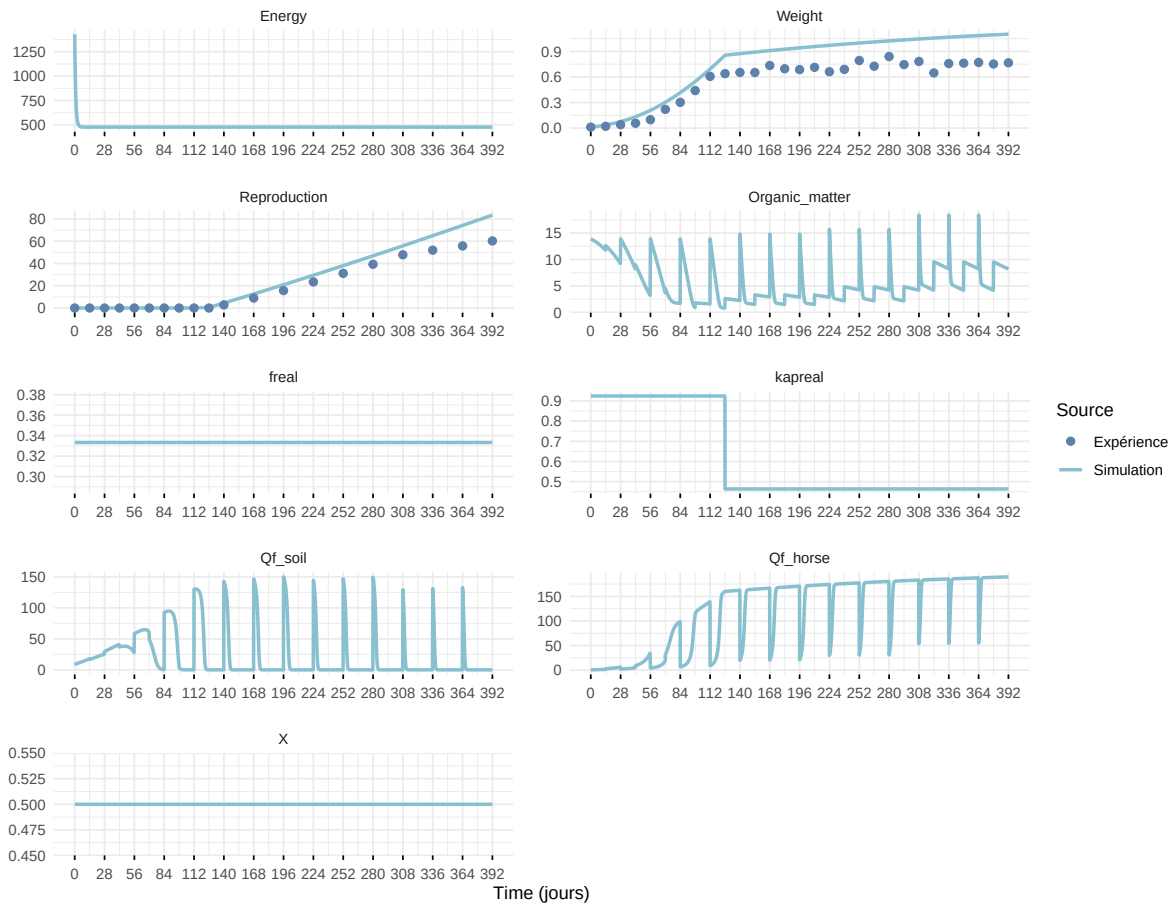
Expérience : Bart2019_XP3



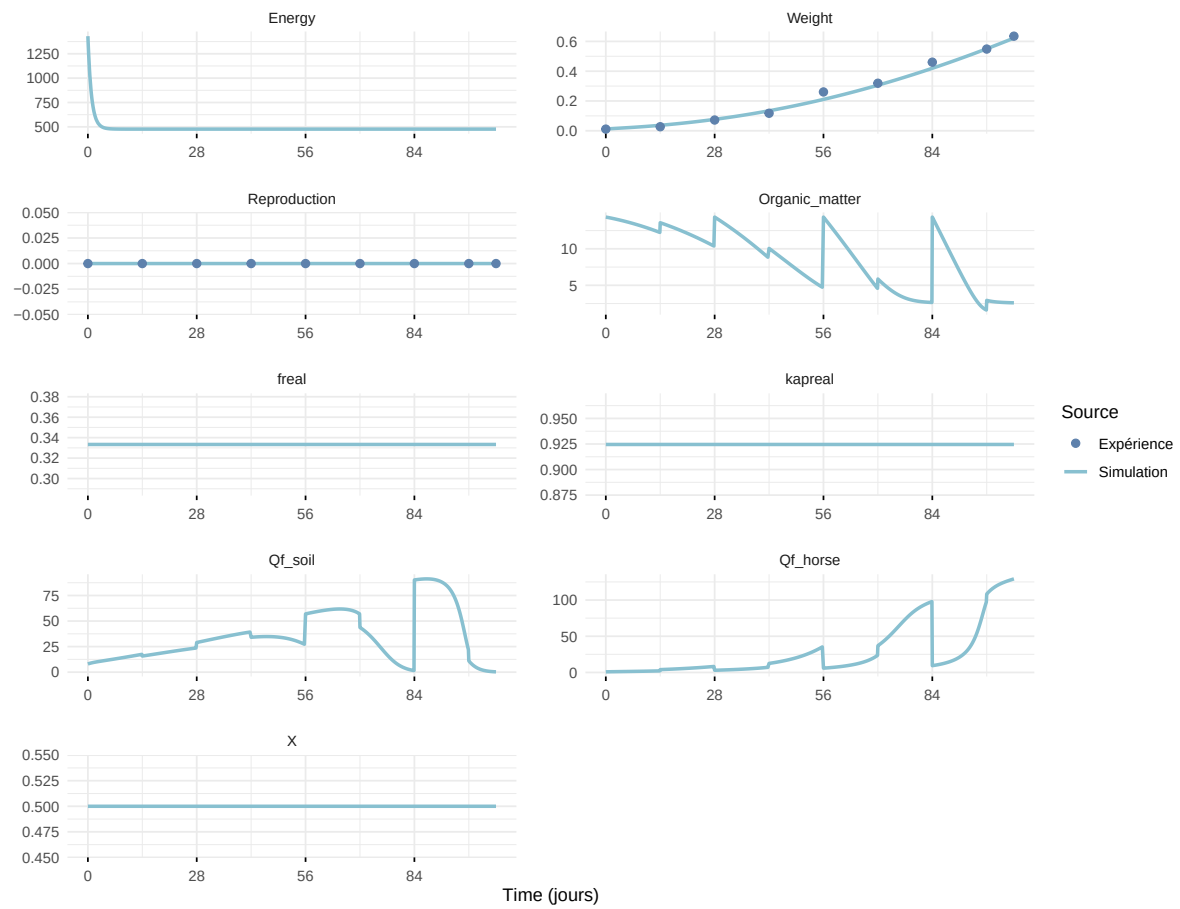
Expérience : Bart2019_XP4_1



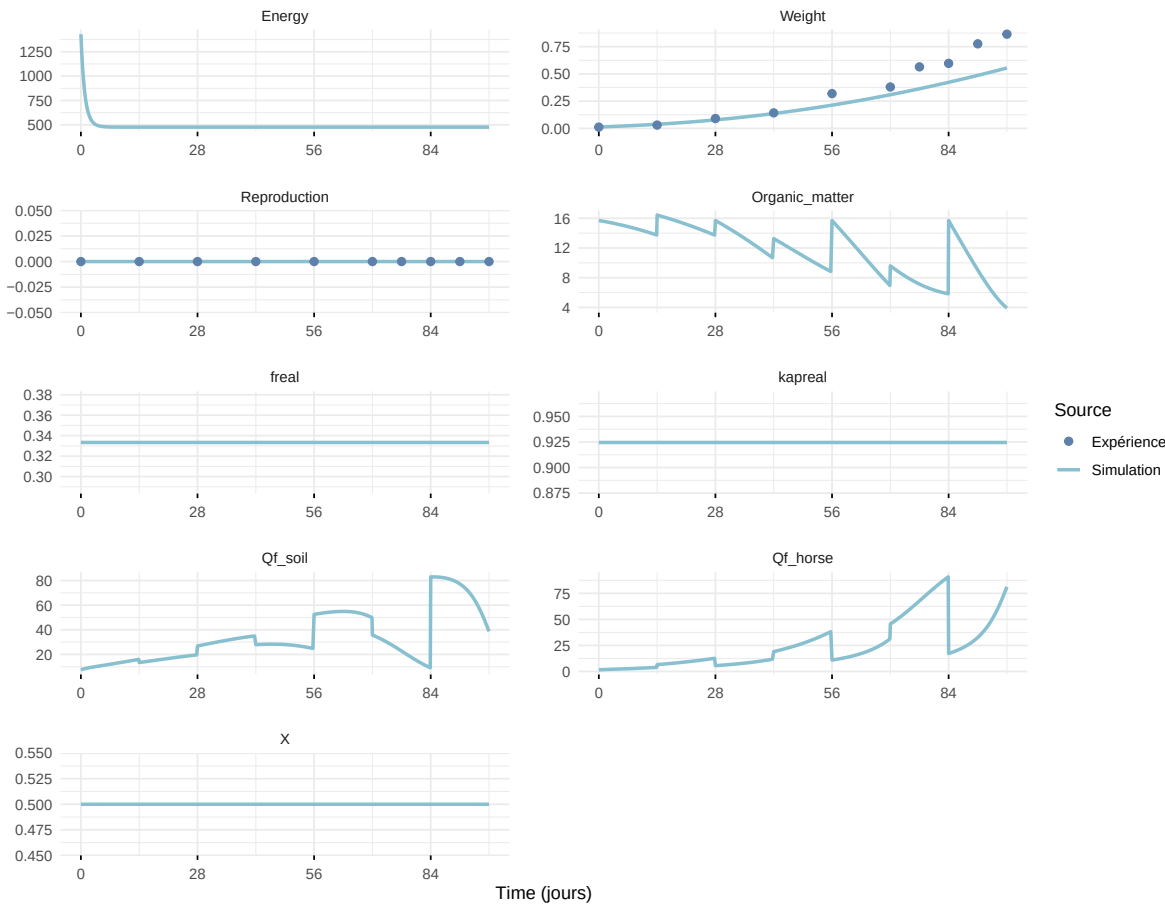
Expérience : Bart2019_XP4_2



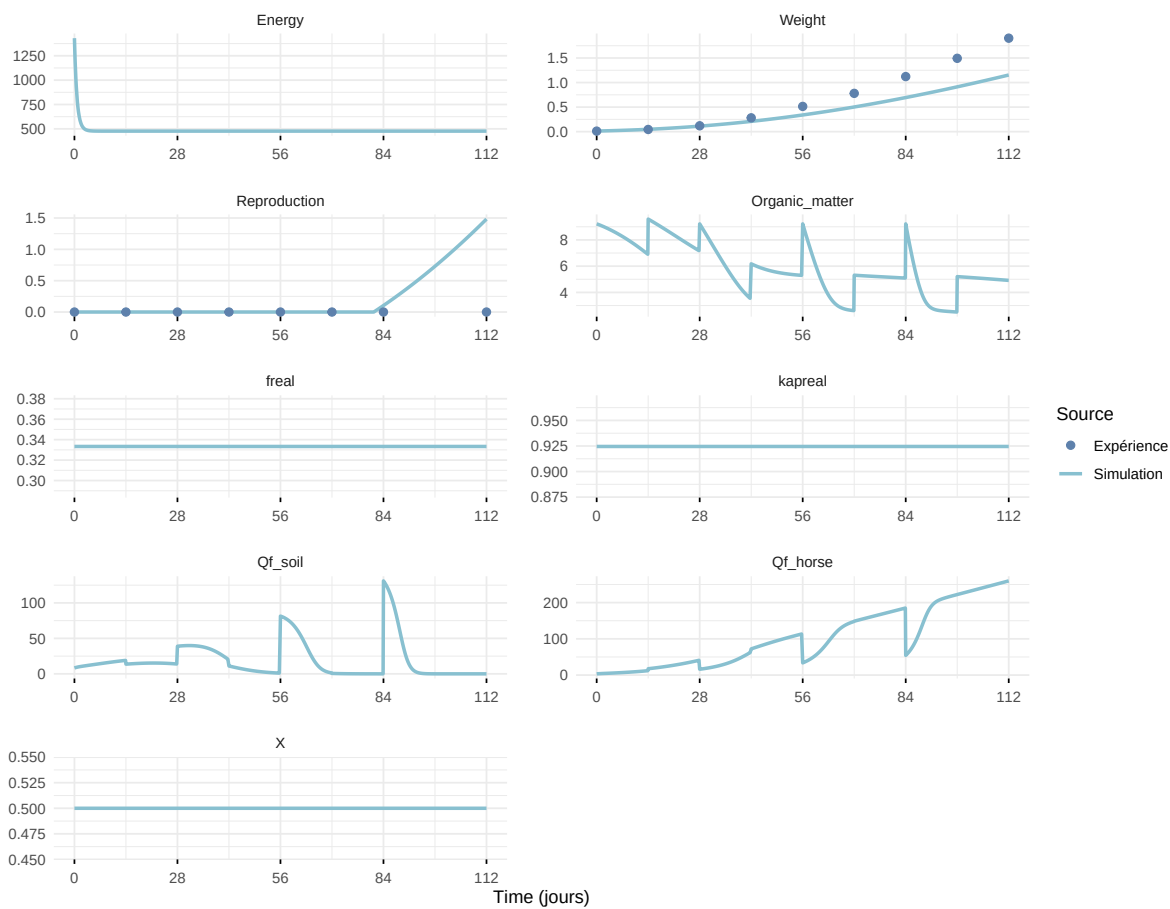
Expérience : Bart2019_XP5



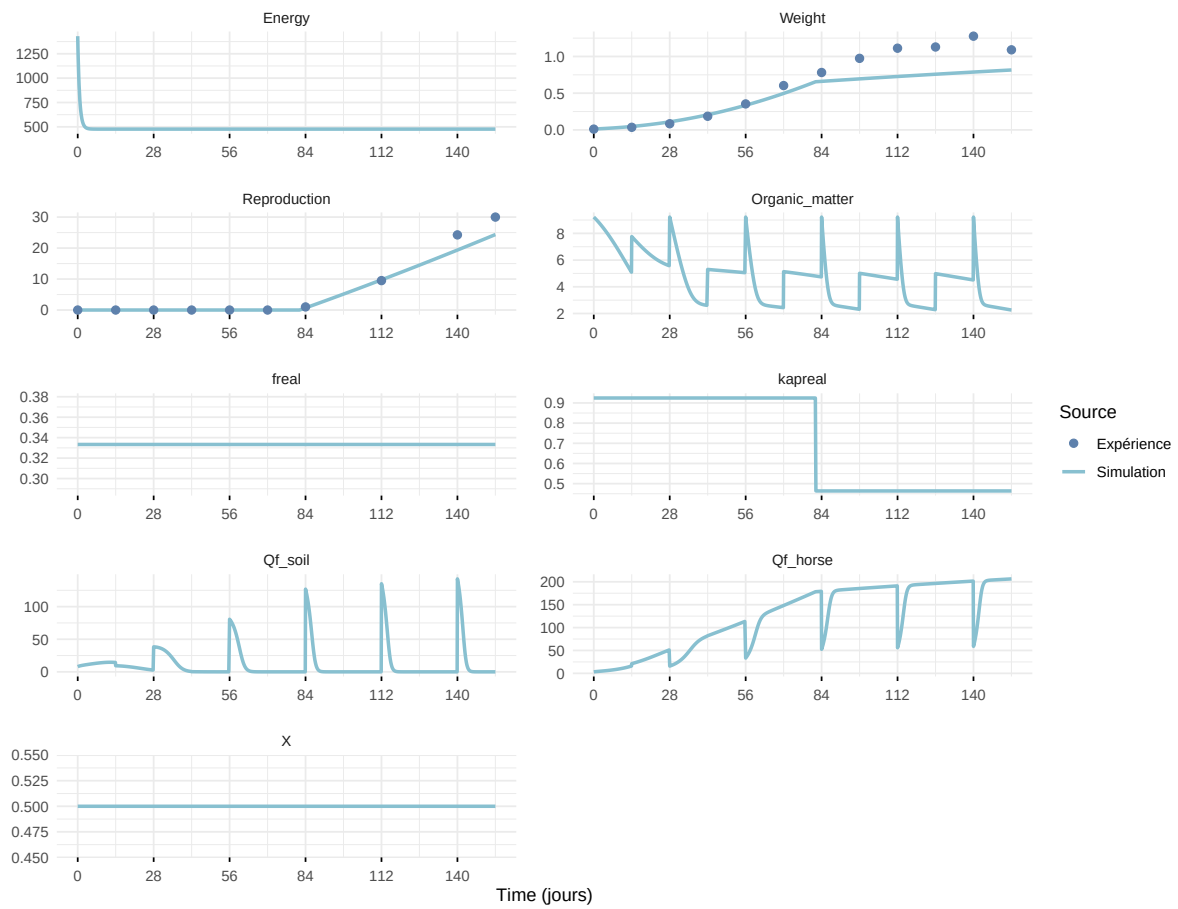
Expérience : Bart2020



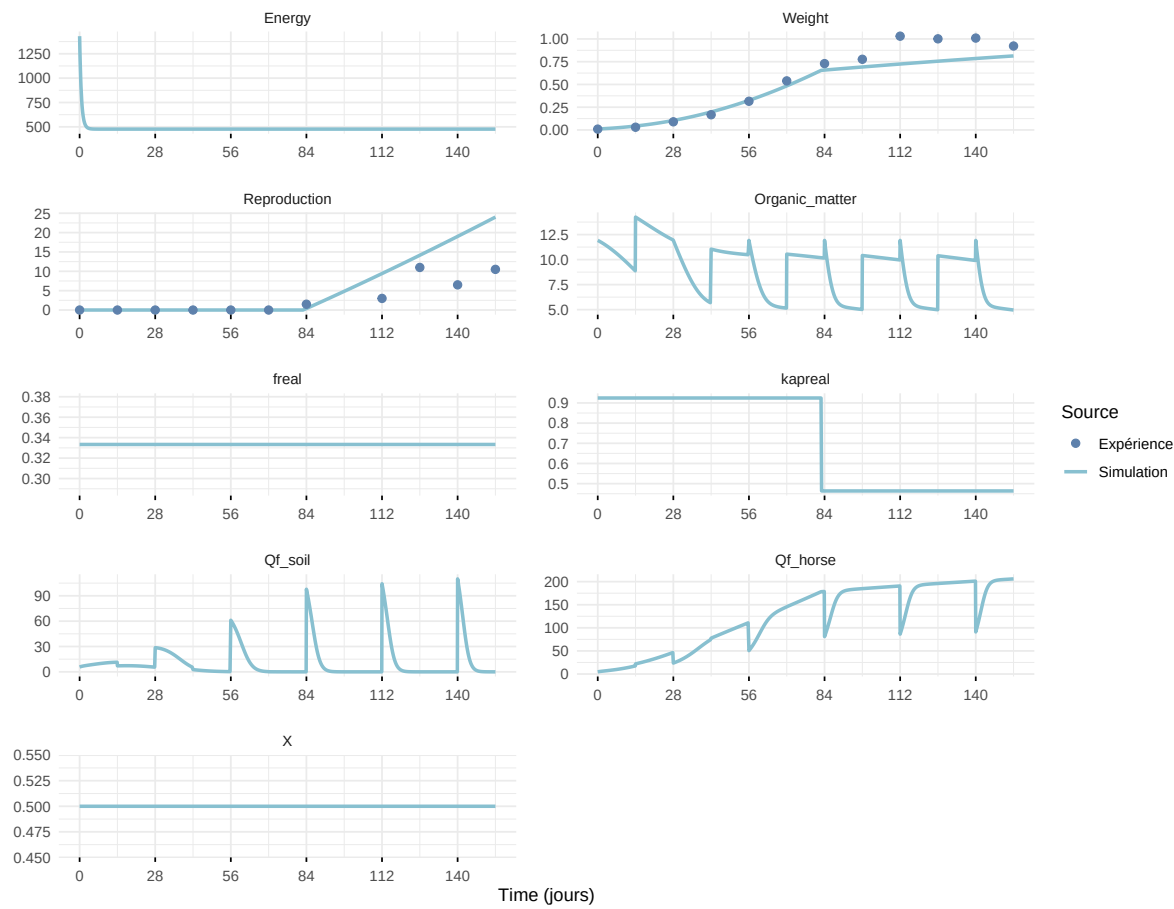
Expérience : Gollot2026_dens_D1



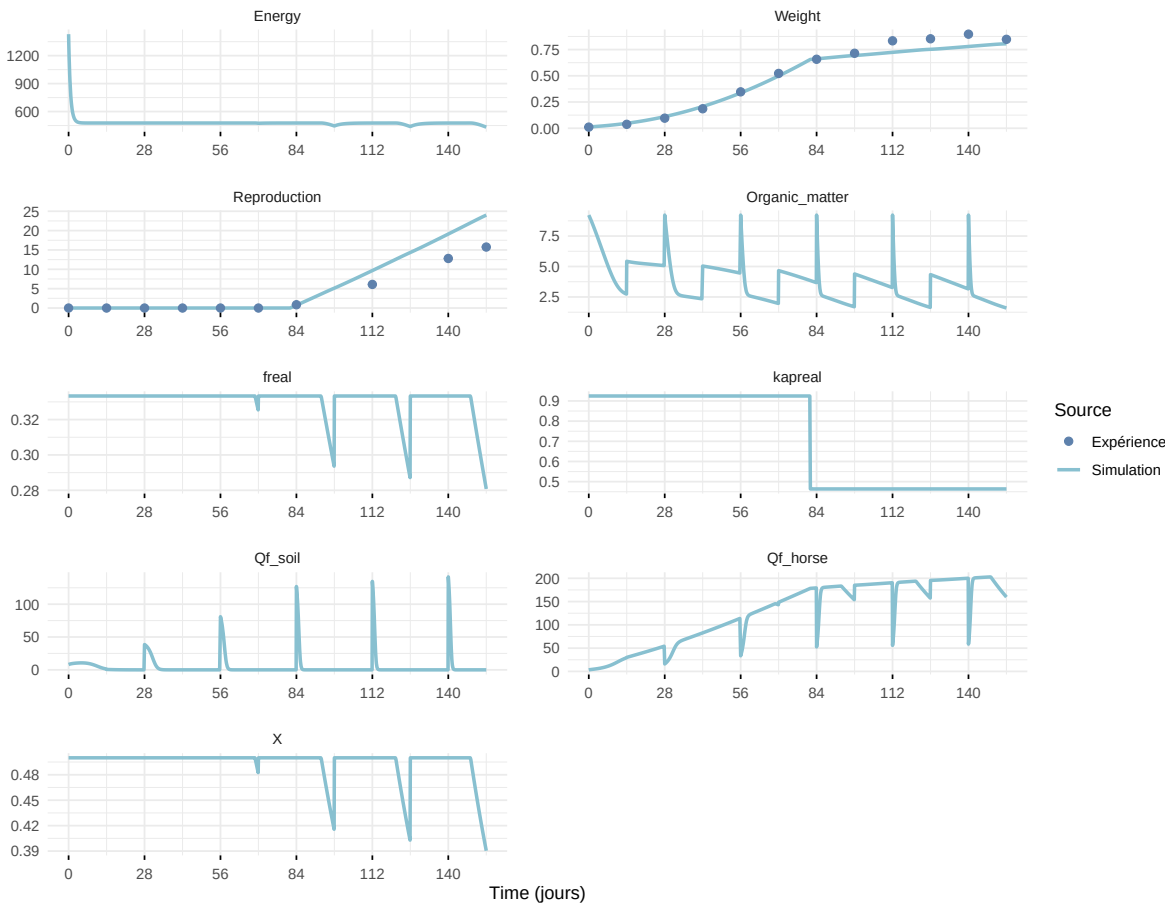
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

